# Souvik Mandol

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**Research Interests:** Machine Learning · LLMs · Human-Computer Interaction · IoT and Systems Security

## Education

**Bangladesh University of Engineering and Technology (BUET)** **2022 – 2026**
Bachelor of Science in Computer Science and Engineering CGPA: 3.31/4.00

## Publications

- **Multi-Model Machine Learning Analysis of Ecological Load Ratio Using Classification, Regression, and Clustering Approaches in EU Countries** *Paper*
  Anik Debnath, Akash Sikdar Sudip, **Souvik Mandol**
  *Proc. 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management, Dhaka, Dec 2025*
  *DOI: 10.46254/BA08.20250232 | Python, Scikit-learn, XGBoost, LightGBM, Optuna, PSO*

  Looked at which national indicators best explain ecological pressure across EU countries, using the Ecological Load Ratio as the target. Compared several tree-based classifiers and regressors, cut a large indicator set down to a handful of key features with PSO and a genetic algorithm, tuned the models with Optuna and checked them with stratified cross-validation. Clustering then grouped countries with similar ecological profiles.

## Research Experience

- **Memory Poisoning Defences for LLM Agents** *2026 – Present*
  *Ongoing Research | In collaboration with Istiaque Ahmed, Graduate Researcher, Osaka Metropolitan University, Japan*

  LLM agents with long-term memory can be misled when someone with access to the user's account plants a false fact over several conversations. We are testing a trust-scoring defence that checks how independently a stored fact is backed up, and comparing it with simpler baselines on several LLMs and a widely used agent-memory library, under a preregistered protocol.

- **Political Bias in LLM-Generated Bengali News Content** *2026 – Present*
  *Ongoing Research | In collaboration with Dr. Sharifa Sultana, Assistant Professor, University of Illinois Urbana-Champaign, USA*

  Looking at whether LLMs shift the political slant of Bengali news when they summarise or rewrite it, a language where this has barely been studied. Helping build a corpus from several Bengali news outlets, annotate it with LLM support, and run human evaluations of model outputs.

- **Age-Aware Neurological Gait Pattern Discovery from Biomechanical Signals: Multi-Phase Stroke Severity Analysis** *2026*
  *B.Sc. Thesis, Dept. of CSE, BUET | Supervisor: Prof. Mahmuda Naznin | Joint work with Tusher Bhomik*

  Stroke changes how people walk, but so do age and walking speed, which makes the effect of the neurological damage itself hard to isolate. Working with a public gait dataset of healthy adults and chronic stroke survivors (kinematic, kinetic and EMG features), we built a multi-phase pipeline that keeps these factors apart: a model of normal ageing to normalise each patient's gait, careful class balancing, and cross-validation designed to avoid data leakage. Removing the age effect noticeably improved classification, and the analysis separated gait markers that mostly reflect walking speed from those that point to the stroke.

## Professional Experience

**Junior Engineer (CSE)**, *BRTC, Bangladesh University of Engineering and Technology* *Aug 2026 – Present*

- Government-funded project with Dhaka South City Corporation to build a locally developed adaptive traffic signal system for Dhaka; live at 4 intersections, scaling to 30.
- Extend and maintain the Go and React/TypeScript platform (MQTT, PostgreSQL/TimescaleDB, WebSocket) that lets operators monitor and control signals from a tablet, with Telegram notifications on every change.
- Built the binary MQTT protocol layer and OTA firmware updates for STM32 Nucleo signal controllers, and the adaptive-mode pipeline that turns Jetson Orin Nano edge-vision traffic data into live, safety-clamped signal plans.
- Set up and run the on-premise server and private fibre network (IP addressing, Docker deployment) linking STM32 controllers, Jetson Orin Nano units, IP cameras and countdown timers, secured with per-device mTLS certificates and topic-level ACLs.

**Software Engineer**, *MerilSoft LLC (Remote, USA)* *Dec 2025 – Apr 2026*

- Built Magazine Works, an AI-powered magazine generation platform, with Next.js, PostgreSQL, AWS and Google Vertex AI.
- Implemented file-handling systems on AWS and integrated AI features through Google Vertex AI Studio.
- Worked on a restaurant system that integrates online delivery platforms such as Uber Eats and DoorDash with a POS solution.

## Projects

- **ResearchQ: AI-Assisted Systematic Literature Review** *Code*

  *Tools: SvelteKit, TypeScript, Drizzle ORM, PostgreSQL, Google Gemini, OpenAlex*

  A self-hosted tool our team built to speed up systematic literature reviews. It searches the OpenAlex graph, screens papers with Gemini, extracts metadata and prepares review outputs, following common protocols such as PRISMA, Cochrane and Kitchenham. Alpha v0.1.0 is archived on Zenodo; a JOSS submission is in preparation.

- **CineRead: Cross-Media AI Recommendation Agent** *Code*

  *Tools: Python, FastAPI, LangGraph, ChromaDB, PostgreSQL, SvelteKit*

  A chat agent that recommends movies, series and books together, in English or Bengali. A LangGraph pipeline works out what the user is asking for, searches TMDB, Open Library and a ChromaDB index of about 940 titles in parallel, links related titles across media and streams ranked answers over SSE. Accounts, saved lists and taste profiles sit in PostgreSQL behind JWT auth.

- **Side-Channel Attack: Website Fingerprinting** *Code*

  *Tools: Python, Flask, PyTorch, Selenium*

  Showed that a website can be identified from CPU cache timing alone, without looking at any network traffic. Collected timing traces with a Selenium-driven browser and trained two 1D CNNs in PyTorch to classify them.

- **GradPilot: AI Platform for Graduate School Applications** *Code*

  *Tools: Next.js, React, Spring Boot, PostgreSQL, Google Gemini* *Demo*

  Helps Bangladeshi students plan graduate school applications: university suggestions, verified mentors, Gemini-based feedback on Statements of Purpose, scholarship listings and a discussion forum. Built as Spring Boot and React microservices.

- **Collaborator: Real-Time Video Meeting Platform** *Code*

  *Tools: Next.js, TypeScript, Supabase, 100ms* *Website*

  A video meeting app with teams, chat, file sharing, recording, and automatic transcripts and summaries. Video runs on 100ms; Supabase handles sign-in, real-time messaging and storage.

## Technical Skills

- **Programming Languages:** Python, Go, C++, TypeScript, JavaScript, SQL, C, Java, HTML, CSS
- **Data Science & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformers, LangGraph, LoRA Fine-tuning, NumPy, Pandas
- **Backend & Systems:** Go, REST APIs, WebSocket, MQTT (Mosquitto), Binary Wire Protocols, mTLS / PKI, Docker, Linux, Tailscale, Private Fibre Network & IP Addressing
- **Embedded & Edge:** STM32 Nucleo and Jetson Orin Nano (host-side integration), Robot Operating System (ROS), Arduino UNO, ATmega32
- **Web Development:** React, Next.js, SvelteKit, Node.js, Express.js, Spring Boot, Laravel
- **Cloud, DevOps & Tools:** AWS, Google Cloud (Vertex AI), Azure, Docker Compose, GitHub Actions, Vercel, Cloudflare R2, Git, Apache JMeter
- **Database Systems:** PostgreSQL, TimescaleDB, Oracle, Supabase, Appwrite

## Certifications

**Coursera:** Neural Networks and Deep Learning · Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization · Structuring Machine Learning Projects

**Kaggle:** Intro to Machine Learning · Intermediate Machine Learning · Pandas

## Languages

Bengali (Native) · English (Fluent)